

Vladislav Makarov

2 March 2003

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EDUCATION

Master's degree — Moscow Institute of Physics and Technology (MIPT) 2025–2027

Phystech School of Applied Mathematics and Informatics

Degree programme: 09.04.01 Informatics and Computer Engineering

Department: Fundamental Methods of Artificial Intelligence

Average grade: 8.47 / 10

Thesis supervisor: Dmitry Yudin, Ph.D. in Engineering

Thesis topic: Development of a monolithic vision-language model for image-based 3D scene graph generation

Bachelor's degree — MIPT 2021–2025

Phystech School of Biological and Medical Physics

Degree programme: 19.03.01 Biotechnology

Department: Bioinformatics and Systems Biology

Average grade: 7.77 / 10

Thesis supervisor: Olga Kudryashova, Ph.D. in Biology

Thesis topic: Analysis of wide-field optical neuroimaging methods using machine learning and computer vision

Presidential Physics and Mathematics Lyceum No. 239 2017–2021

WORK EXPERIENCE

Intelligent Transport Laboratory, Center for Cognitive Modeling, MIPT August 2025 — Present

Position: Intern

Scientific supervisor: Dmitry Yudin

Federal Center of Brain Research and Neurotechnologies (FCBRN), FMBA of Russia September 2024 — July 2025

Position: External research staff member

Scientific supervisor: Olga Kudryashova

National Medical Research Center for Children's Health September 2024 — July 2025

Position: External research staff member

RESEARCH PROJECTS

Sber June 2026 — August 2026

Development of a methodology for automatically creating training data for vision-language models that generate humanoid robot actions

Sber August 2025 — December 2025

Development of intelligent methods for constructing and updating knowledge maps and their application to mobile manipulator localisation

RESEARCH PUBLICATIONS

- Makarov V.**, Gizetdinov M., Yudin D. SceneGraphVLM: Dynamic Scene Graph Generation from Video with Vision-Language Models. *arXiv:2605.13667*, 2026. [arXiv](#)
- Pinkin E., Kartashov G., Kalmykov M., **Makarov V.**, Melekhin A., Yudin D. GraphSeqLoc: Image Sequence-Based Place Recognition via Scene Graphs and Temporal Re-Ranking. *SSRN 7422535*, 2026. [SSRN](#)

CONFERENCES

67th All-Russian Scientific Conference of MIPT

April 2025

Track: Biological and Medical Physics

Session: Bioinformatics and Systems Biology

Presentation title: Bioinformatic analysis of wide-field optical neuroimaging data to investigate neurovascular changes in ischaemic stroke

Result: Winner

All-Russian Conference “Neurocampus 2025”

April 2025

Interdisciplinary approaches to the study and treatment of nervous system diseases

Format: Poster presentation

Presentation title: Analysis of neurovascular changes in ischaemic stroke using wide-field optical neuroimaging data

PROFESSIONAL DEVELOPMENT

Advanced Machine Learning Methods

February 2023 – December 2024

MIPT Digital Department – Diploma of Professional Retraining

TEACHING

Teaching Assistant – Phystech@DataScience, MIPT

September 2024 – June 2025

MIPT · [Statistics and Machine Learning for Applied Problems](#)

Led tutorials in statistics and machine learning

HACKATHONS

SIBUR – “Element 119” AI in Chemistry Competition

April 2025

Task: Prediction of the lipophilicity of organic molecules

Researcher · ML Engineer · Participating team · [GitHub](#)

BostonGene

December 2024

Task: Analysis of immune repertoires in response to a pathogen by matching T-cell receptor CDR3 regions to known sequences in a database

Researcher · ML Engineer · **Winning team** · [GitHub](#)

MIPT Digital Department – Gazprom Neft

November 2023 – December 2023

Task: Development of a web service for the automatic grouping of procurement requests for materials and equipment into lots

Team Captain · ML Engineer · Participating team

CASE COMPETITIONS

Phystech Business Solutions – Yakov and Partners

November 2023 – December 2023

Task: Tripling the number of unique patients attending annual check-ups across the company’s clinic network within five years

High-tech business · Researcher · Data Analyst · Participating team · [Project materials](#)

Phystech Business Solutions – Kept

November 2022 – December 2022

Task: Development and justification of initiatives to optimise the business functions of an oil-refining holding company without changing production processes, with the aim of improving company performance

Researcher · Data Analyst · **Finalists, internship invitation** · [Project materials](#)

Phystech Business Solutions – Zerion

April 2022 – May 2022

Task: Development of Web 3.0 solutions for secure P2P services that enhance trust, transparency and accessibility in economic interactions

Researcher · Data Analyst · **Finalists, internship invitation** · [GitHub](#)

INTERNSHIPS

Laboratory of Molecular Medical Diagnostics, MIPT

July 2022 – August 2022

Intern · Developed professional skills in mass spectrometry

Infochemistry Scientific Center, ITMO University

November 2019

Intern · Developed professional skills in infochemistry and materials science

KEY SKILLS

Languages and Frameworks	Python, PyTorch, Lightning, NumPy, Pandas, SciPy, scikit-learn, OpenCV, scikit-image, Seaborn, Matplotlib
Tools	Linux, Bash, Git, Docker, Jupyter, Isaac Sim, Rerun, CVAT, Codex, Cursor
Specialisation	Deep Learning, Computer Vision, Vision-Language Models (VLM), 3D Scene Graphs
Language Proficiency	Russian — Native · English — B1

REFERENCES

Dmitry Yudin

MIPT, Ph.D. in Engineering, Head of the Intelligent Transport Laboratory, Center for Cognitive Modeling and Scientific and Design Bureau of Computing Systems

Elizaveta Chefanova

MIPT, Team Lead, Product Analytics, Yandex

Olga Kudryashova

FCBRN, FMBA of Russia, Ph.D. in Biology, Head of the Bioinformatics and Omics Technologies Group

Radoslav Neychev

MIPT, Ph.D. in Computer Science, Co-developer of the master's programme "Modern Methods of Artificial Intelligence", Founder of girafe-ai

Daria Novikova

Saint Petersburg State University, Senior Lecturer, Department of Foreign Languages for Physics, Mathematics and Chemistry